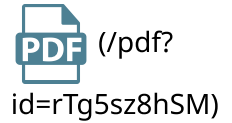




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CausalMix: Defining and Benchmarking Hybrid Causal Discovery



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04 May 2026 (modified: 28 May 2026) NeurIPS 2026 Evaluations and Datasets Track Submission
 Evaluations and Datasets Track, Senior Area Chairs, Area Chairs, Reviewers, Authors
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Keywords: causal discovery, LLM-Assisted Causal Discovery

TL;DR: We formalize mixed-information causal discovery and introduce CAUSALMIX, a benchmark revealing that LLMs recover causal graphs mostly from variable names alone, with data adding only weak, format-sensitive gains well short of classical methods.

Abstract:

Classical causal discovery recovers structure from numerical data, while large language models (LLMs) carry semantic priors about variables and mechanisms from pretraining. Research that combines the two is nascent and lacks a unified problem definition or controlled evaluation protocol: existing benchmarks cover the data-only and semantic-only endpoints but leave the middle ground untested. To bridge the gap, in this work, we make two contributions. First, we formalize hybrid causal discovery and release CausalMix, a simulation and evaluation suite that generates matched observational and interventional datasets from Bayesian-network graphs and turns the regime into controlled attribution tests by varying semantic cues, data budget, and data representation while fixing the graph. The suite reports a semantic-only reference (what variable names alone recover) and a data-only reference (what classical algorithms recover from the same samples), so any gain from a hybrid method can be attributed to its inputs. Across ten LLMs on four reference graphs, our controlled evaluation shows that zero-shot prompting recovers substantial structure from names alone, while adding numerical evidence yields weak and format-sensitive gains that fall well short of classical baselines on the same data. Second, as a preliminary step beyond this baseline, we introduce a post-training procedure that uses ground-truth graphs as verifiers to generate observational and interventional supervision for LLM fine-tuning. The fine-tuned model shows early evidence of combining semantic priors with numerical data, with consistent gains on several small graphs, although a gap to classical methods remains. We release CausalMix as an open-source toolkit to support a more rigorous evaluation protocol for LLM-assisted causal discovery.

Review Mode: Double-blind (default; anonymized submission)

Dataset Submission: This submission includes a dataset.

Croissant File: json (/attachment?id=rTg5sz8hSM&name=croissant_file)

Dataset URL: <https://huggingface.co/datasets/mixcausalbench/anonymous-data>
<https://huggingface.co/datasets/mixcausalbench/anonymous-data>

Dataset Large URL: <https://huggingface.co/datasets/mixcausalbench/anonymous-data>
<https://huggingface.co/datasets/mixcausalbench/anonymous-data>

Code Submission Justification: Uploaded as Supplementary Material

Supplementary Material: zip (/attachment?id=rTg5sz8hSM&name=supplementary_material)

Primary Area: Language and multimodal language models (e.g., text generation, summarization, VQA)

Contribution Type: Evaluation methodology and metrics, e.g., evaluation protocols, metric design and validation, experimental methodology, statistical evaluation methods, comparisons of evaluation designs, refining evaluation setups.

LLM Usage:  Editing (e.g., grammar, spelling, word choice)

Author Acknowledgements:  **CONTRIBUTION ACKNOWLEDGEMENT:** Please confirm that the main paper clearly articulates the evaluative role of the contribution, including what claims it supports, under which assumptions these claims hold, and its limitations. For datasets, the submission should describe how the dataset is intended to be used meaningfully in evaluative practice rather than as an endpoint., **CHECKLIST CONFIRMATION:** In order to improve the rigor and transparency of research submitted to and published at NeurIPS, authors are required to complete a paper checklist and include it in the paper PDF. See <https://neurips.cc/public/guides/PaperChecklist> (<https://neurips.cc/public/guides/PaperChecklist>) and the provided 2026 sample LaTeX file. Papers not including the checklist in the paper PDF will be desk rejected. Please confirm that a paper checklist is included in the paper PDF., **RESPONSIBLE REVIEWING:** We acknowledge that every author on the program committee is obliged to conduct responsible ACing / reviewing according to the policy designed by the PC Chairs and relevant members of the organizing committee. Failure to do so may result in desk rejection or other serious consequences., **ACADEMIC INTEGRITY:** See the NeurIPS Code of Conduct (<https://neurips.cc/public/CodeOfConduct> (<https://neurips.cc/public/CodeOfConduct>)) and the NeurIPS policy of Academic Integrity in Research Paper Submissions and Reviews (<https://neurips.cc/Conferences/2026/ReviewerIntegrity> (<https://neurips.cc/Conferences/2026/ReviewerIntegrity>)). I acknowledge that I have read and commit to adhering to the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy., **DECLARATION:** I confirm that the information provided in this form is accurate.

Reviewer Nomination:  Yuen Chen (/profile?id=~Yuen_Chen1)

Submission Number: 3307

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Meta Review of Submission3307 by Area Chair Kf9w

Meta Review by Area Chair Kf9w  21 Jul 2026, 10:20 (modified: 23 Jul 2026, 12:23)

 Senior Area Chairs, Area Chairs, Authors, Reviewers Submitted, Program Chairs, Area Chair Kf9w

 Revisions (/revisions?id=ZRvSqPyi37)

Metareview:

Scientific Contribution. CausalMix formalizes hybrid causal analysis, in which models receive both semantic information about variables and numerical observations or interventions. The controlled evaluation distinguishes between "names-only," "data-only," and combined scenarios to determine whether LLM performance relies on semantic prior knowledge or the actual use of numerical evidence. Experiments suggest that LLMs reconstruct substantial graph structures from variable names, whereas numerical data offer only limited and format-dependent benefits. The paper also presents a preliminary, verifier-based fine-tuning approach.

Strengths. The controlled attribution model addresses a key methodological issue in LLM-based causal analysis: high graph reconstruction scores do not necessarily prove that a model utilized the provided data. Comparing scenarios involving purely semantic information, hybrid data, and classical data is useful and potentially widely applicable. Comparisons with classical causal analysis algorithms and a generally transparent discussion of limitations represent further strengths.

Main Criticisms. The primary criticism is that the key conclusions rely largely on four small, widely known graphs containing 5–11 nodes. These graph structures and variable names may already be present in the model's pre-training data; consequently, the strong results observed in the "names-only" scenarios may stem partly from rote memorization. Thus, the findings do not demonstrate that the observed behavior generalizes to larger, less common, or newly generated causal structures.

Furthermore, the fine-tuning experiment lacks sufficient validity: the training and evaluation phases utilize overlapping graph families, meaning that improvements could result from memorizing structures, output formats, or the verifier's preferences rather than from an improved integration of semantic and numerical evidence. A graph-disjoint evaluation is required.

After all, many experimental cells contain only a few realizations, and some of the reported differences are small. More robust uncertainty estimates and repeated runs are necessary before these effects can be interpreted as robust. Furthermore, model coverage is lower than expected for a benchmark-based claim, and some key concepts and methodological details require clearer definition.

Add: **Author AC Confidential Comment**

Review of paper 3307

Official Review by Reviewer svky 📅 25 Jun 2026, 23:04 (modified: 23 Jul 2026, 11:25)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer svky

📄 Revisions (/revisions?id=O9uFOWqWrT)

Summary:

The paper studies hybrid causal discovery methods (LLM + statistical methods) in three aspects: (1) formally define a hybrid causal discovery method; (2) introduce an evaluation frame that checks the effectiveness of LLM and statistical-based components separately; (3) fine-tune an LLM to complete causal discovery tasks. Additionally, the work introduces a simulator to generate benchmarks for hybrid causal discovery method evaluation. The work evaluates hybrid methods based on several LLM models and several statistical-based causal discovery methods by their evaluation frame, concluding that pure LLM-based methods perform well on several small datasets, while statistical-based methods are stronger for larger-scale tasks. They also show that task-driven fine-tuning can significantly improve the performance of LLM-based methods.

Contribution Type Check: Yes

Strengths:

The paper formalises a frame for the hybrid causal discovery methods to evaluate the performances of the statistical-based component and the semantic-based component separately. Such an evaluation process would be helpful to improve the hybrid causal discovery methods.

Weaknesses:

1. The paper's novelty is limited. It may try to shape the evaluation frame as a standard for hybrid methods' evaluation, however, many of the components have been used in existing works. The hybrid method's definition is also obvious. Fine-tuning LLMs to handle causal discovery tasks also exists in previous works.
2. The presentation needs improvement. I could not clearly understand every detail of the method, e.g., many terms do not have an explanation. To name a few: "semantic cue", "matched attribution problem". I believe the authors need to significantly revise the methodology to explicitly describe the components instead of only provide intuition with such vague terms.
3. Another concern is the paper's unfair attitude towards the semantic component (e.g., Lines 104-112). My understanding is that the authors tend to believe that the numerical data is the gold standard for causal discovery, but the semantic evidences are not that reliable. However, in practice, a numerical dataset also suffers from bias, and in contrast, semantic evidence may help debias the result. For instance, LLMs provide suggestions based on a vast domain research training dataset, potentially more reliable than a single dataset.
4. The authors use several toy datasets, especially, the Asia, Earthquake, and Cancer. These datasets may be out of date, and moreover, some of them contain semi-synthetic data, hence not suitable for evaluating LLM-based

methods. The paper also shows that a simple strategy to use pure LLM-based methods can achieve good performances. This may be because of LLMs' memory of the corresponding datasets.

Ethical Concerns: No or very minor ethics concerns only

Reproducibility: Yes

Dataset Assessment: NA - no dataset included

Limitations:

I would recommend that the authors add experiments on the latest benchmarks to test the performances.

Questions:

See the weakness section.

Rating: 2: Reject: For instance, a paper with technical flaws, weak evaluation, inadequate reproducibility and incompletely addressed ethical considerations.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Paper Formatting Concerns:

No concern.

LLM Policy: Yes

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

Add: **Author AC Confidential Comment**



Rebuttal by Authors

Edit ▼



Rebuttal by Authors 📅 27 Jul 2026, 19:53 👁 Program Chairs, Authors

Rebuttal:

Thank you for recognizing the value of separately evaluating semantic and statistical evidence. To address the scale and familiarity concerns, we added larger and newly generated graphs plus a name-data conflict control; recovery remains weak, so we narrow rather than broaden our claims. We will also clarify the paper's contribution and terminology.

W1 — Novelty

CausalMix is an evaluation and dataset contribution, not a new causal-discovery algorithm. We do not claim novelty for supervised fine-tuning (SFT), replacing variable names with neutral labels, prompting with names but no numerical data, representation comparisons, or classical baselines individually. Our contribution is a matched attribution protocol: it changes variable names, numerical evidence, or representation one at a time while holding the remaining factors fixed. This identifies whether performance comes from names, data, representation, or their interaction—information that aggregate graph scores cannot provide. CausalMix supplies the datasets, simulator, prompt builders, classical baselines, and shared scoring needed to make these comparisons reproducible. We will center this contribution in the paper and present post-training as a case study of the benchmark, not as a new training method.

W2 — Presentation and terminology

- A **semantic cue** is text associated with a variable, principally its original name and description. Anonymization replaces it with a neutral label such as **X1**.
- A **matched attribution test** changes one factor—names, numerical evidence, or representation—while fixing the graph, data realization and seed, observation/intervention budget, variable order, model settings, required input/output format, and scoring.

For example, a real-name/anonymized Sachs pair contains identical samples and differs only in labels such as Raf and Mek. Data/no-data pairs isolate responsiveness to numerical evidence; summary/tabular pairs compare statistical summaries with row-wise samples. We use **name-mediated support** to mean

performance enabled by variable names, whether it comes from exact benchmark recall or broader knowledge about the named variables. Anonymization cannot distinguish those two sources. We will add these definitions and this matched example to the revised methodology.

W3 — Treatment of semantic and numerical evidence

Our wording may have unintentionally implied a hierarchy between label semantics and data. We shall revise the text appropriately to address this unintended hierarchy. Semantic knowledge may encode expertise or misleading associations; numerical evidence depends on causal, sampling, and measurement assumptions. CausalMix measures both sources and does not treat data-only methods as gold standards. In controlled cells, all methods are scored against the known generating directed acyclic graph (DAG); the classical causal-discovery methods PC, GES, and ENCO are baselines under their assumptions. We will revise the relevant language.

W4 — Dataset familiarity, age, and scale

The familiarity of Asia, Earthquake, Cancer, and Sachs increases the risk of pretraining recall. Their known DAGs and simulators nevertheless enable exact control over observational samples, intervention targets, samples per intervention, and seeds, allowing real-name and anonymized conditions to use identical data and budgets. We group both exact graph recall and general knowledge about named variables as **name-mediated support**, rather than data-driven causal discovery.

To address limited graph scale and possible reliance on familiar benchmark structures, we added 20–37-node evaluations, including the newly generated Chain-25 and Jungle-25, plus degree-preserving rewired Child variants that place familiar names in conflict with new numerical evidence. These controls probe, but cannot eliminate, pretraining familiarity.

For these evaluations, **Base** denotes Qwen3-4B-Thinking-2507 before our fine-tuning. **Format SFT** trains the model to produce structured responses containing gold graph targets; **reasoning SFT** then continues supervised training with gold answers, concise rationales, and teacher-generated rationales. We evaluated the model after both SFT phases on 20 paired realizations per comparison: 4 fixed column-order seeds × 5 independent data draws per order. Within each realization, the same draw is shared across label conditions. The table reports data-present results after format SFT + reasoning SFT. **Validity** is the percentage of outputs satisfying the structured-output contract and parseable at the expected graph size; acyclicity is evaluated separately. Directed-edge **F1** measures recovery of the true directed edges, with invalid outputs assigned F1=0. Child and Alarm are established 20- and 37-node graphs; Chain-25 is a directed chain; and Jungle-25 is a denser hierarchical DAG with connections spanning up to two levels.

Because an edge list represents a set, exact repeated copies of the same directed edge are collapsed before scoring; they neither add an edge nor invalidate the graph. Self-loops, unknown endpoints, malformed edges, and truncations remain invalid. All values below use this scoring rule and aggregate 20 runs per cell. ENCO uses the same 20 datasets and numerical budget (1,000 observational samples and 10 samples per intervention):

Graph	ENCO F1	LLM original: validity / F1	LLM anonymous: validity / F1
Child (20)	0.652	100% / 0.070	100% / 0.104
Chain-25	0.609	90% / 0.049	100% / 0.069
Jungle-25	0.703	95% / 0.044	100% / 0.095
Alarm (37)	0.677	60% / 0.022	90% / 0.031

Recovery remains weak despite substantially higher parse validity. The corresponding data-use gain (F1 with data minus F1 without data, invalid outputs scored 0) is:

Graph	Original-label gain	Anonymous-label gain
Child (20)	0.000	+0.005
Chain-25	0.000	0.000

Graph	Original-label gain	Anonymous-label gain
Jungle-25	+0.003	0.000
Alarm (37)	−0.133	0.000

Takeaway: ENCO reaches F1=0.609–0.703, confirming that the simulated numerical signal is recoverable under a matched data budget. The LLM remains at 0.022–0.104 despite 90–100% validity on most cells, and its data-use gain is approximately zero except for a negative original-label effect on Alarm; that effect partly reflects Alarm's lower 60% data-present validity because invalid outputs receive zero. Chain-25 and Jungle-25 are synthetic categorical graphs with neutral variable labels. We also created five Child variants that retain the original names and each variable's in- and out-degree while replacing 12–16 of the graph's 25 edges and preserving acyclicity. Because data come from the rewired graphs, these tests place familiar names in conflict with numerical evidence. These controls test unfamiliar structures and name–data conflict, but cannot rule out all pretraining effects or replace evaluation in a new scientific domain. We therefore limit our conclusions accordingly.

Thank you again for the thoughtful feedback. We will use these suggestions to clarify CausalMix's contribution, terminology, treatment of semantic and numerical evidence, and empirical scope in the revision.

Interesting ablation study with a flawed SFT experiment design

Official Review by Reviewer mUVt 24 Jun 2026, 19:02 (modified: 23 Jul 2026, 11:25)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer mUVt
Revisions (/revisions?id=5KvIDX77Z6)

Summary:
The paper studies hybrid causal discovery. In this setting, a method uses both variable meanings and numerical data to recover a causal graph. The authors argue that current LLM evaluations often mix these signals. A model may score well because it knows what variable names imply, not because it used the data. The main contribution is CAUSALMIX, a benchmark suite that controls this issue. It compares names-only prompts, real-name prompts with data, anonymized prompts with the same data, different data formats, and classical data-only baselines. The authors evaluate several LLMs on graphs such as Cancer, Earthquake, Asia, and Sachs. They find that LLMs recover substantial structure from names alone. Adding data gives limited and format-sensitive gains. Classical methods remain stronger when the statistical setting fits their assumptions. The paper also includes a preliminary fine-tuning experiment for Qwen3-4B. This improves output validity and some graph-recovery scores.

Contribution Type Check: Yes

- Strengths:**
1. The paper makes a clear and useful contribution to LLM-based causal discovery. Its main strength is the attribution strategy. The authors show that a high graph-recovery score is not enough, since an LLM may recover edges from variable names rather than from the supplied data. The work is significant because it exposes a key limitation of current LLM-CD methods. The reported gains often appear to come largely from semantic priors rather than robust use of observational or interventional data. This is an important diagnostic for the field.
 2. The empirical evaluation is broad. The authors test many baselines, including LLM-based methods and conventional causal discovery methods such as PC, GES, and ENCO. This makes the comparisons more informative than evaluating LLMs in isolation.
 3. The authors are appropriately transparent about limitations. In particular, they acknowledge the limited statistical power in parts of the study and treat some findings as preliminary rather than definitive. This makes the contribution more credible.

Weaknesses:

1. The SFT experiment has a weak train/test split. The training data uses the same graph families that are later evaluated. Stage 1 includes Cancer, Earthquake, Asia, and Sachs. Stage 2 includes Earthquake, Asia, and Sachs. The evaluation is again on Cancer, Earthquake, Asia, and Sachs. This makes it hard to tell whether the model

learned to use numerical evidence, or simply learned the graph structures, output format, and verifier preferences. A graph-disjoint evaluation would be needed to support stronger claims.

- 2. The limitation on statistical power is real. Many repeated cells use only five data realizations. The bootstrap intervals are useful, but the authors state that they are descriptive rather than high-powered tests. This weakens claims about small F1 differences, such as data gains, format effects, anonymization drops, and model comparisons. The broad patterns are still useful, but small effects should be interpreted cautiously.
- 3. The studied causal graphs are small and well-known. Cancer and Earthquake have only five nodes, Asia has eight, and Sachs has eleven. These are common benchmark graphs, so they may already be familiar to LLMs through pretraining. It is therefore somewhat expected that GPT-style models can recover structure from variable names alone. The paper would be stronger if it tested whether the same pattern holds on larger, less popular, or newly constructed causal graphs.

Ethical Concerns: No or very minor ethics concerns only

Reproducibility: Yes

Reproducibility Comments:

The authors provide the anonymous codebase (through complementary material in openreview) and dataset (hf). I did not run them manually but everything seems pretty organized and straightforward to reproduce with API keys.

Dataset Assessment: Yes

Dataset Comments:

See above.

Limitations:

Overall, I would say the limitations are honestly discussed, but not fully resolved. The paper is more of a benchmark and evaluation-protocol contribution. It is weaker as (formal) evidence about current LLMs’ general ability to use causal data or about the effectiveness of post-training.

Questions:

- 1. Could the authors clarify the intended claim of the SFT experiment? Since training and evaluation use the same graph families, gains may reflect simply memorization. A small leave-one-graph-out evaluation would make this much stronger.
- 2. The names-only result may be affected by the popularity of Cancer, Earthquake, Asia, and Sachs. Can the authors test less common or newly constructed named graphs? This would help separate semantic reasoning from benchmark memorization.
- 3. The main matched experiments are on small graphs. Can the authors add a lightweight matched evaluation on a larger graph, even with fewer models or summary-only prompts? This would clarify whether the attribution findings scale.

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Paper Formatting Concerns:

N/A

LLM Policy: Yes

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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Author AC Confidential Comment



Rebuttal by Authors

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Rebuttal by Authors 27 Jul 2026, 19:54 Program Chairs, Authors

Rebuttal:

We appreciate the reviewer’s positive assessment and concrete suggestions. We completed strict held-out-family training, a 30-realization replication, and tests on larger and newly generated graphs. The held-out results show transferable output reliability but mixed numerical-evidence gains, while larger-graph recovery remains weak; we narrow the claims accordingly.

W1 / Q1 — Post-training ablation and held-out-graph evaluation

We used strict leave-one-graph-out (LOGO) training, so train and test graph families do not overlap: each graph is evaluated only by models whose training data exclude it. **Base** denotes Qwen3-4B-Thinking-2507 before our fine-tuning. **Format supervised fine-tuning (format SFT)** first trains the model to produce structured responses containing gold graph targets. Starting from that checkpoint, **reasoning SFT** continues training with gold answers and rationales. Each of Cancer, Earthquake, Asia, and Sachs was removed from both phases, and every fold was trained with seeds 42, 314, and 2718.

Output validity is the percentage of responses satisfying the structured-output contract and parseable at the expected graph size; acyclicity is evaluated separately. Exact repeats of the same directed edge are collapsed before scoring. Aggregate validity averages the real-name, anonymous (neutral-label), data, and no-data conditions:

Held-out graph	Base	After format SFT	After format SFT + reasoning SFT
Asia	71.3%	100%	100%
Cancer	86.3%	100%	100%
Earthquake	76.3%	100%	100%
Sachs	73.8%	100%	100%

Directed-edge **F1** measures recovery of the true directed edges, with invalid outputs assigned F1=0. We define **data-use gain** as F1(with data) – F1(without data). The next table reports how reasoning SFT changed this gain relative to format SFT, with 95% bootstrap CIs over 60 paired predictions per graph and naming condition: 20 data pairs (4 fixed column-order seeds × 5 independent data draws per order) × 3 training seeds.

Held-out graph	Real names: reasoning SFT effect [95% CI]	Anonymous: reasoning SFT effect [95% CI]
Asia	−0.071 [−0.116, −0.028]	+0.028 [+0.003, +0.054]
Cancer	+0.003 [−0.056, +0.061]	+0.004 [−0.056, +0.069]
Earthquake	−0.025 [−0.074, +0.027]	+0.037 [+0.012, +0.064]
Sachs	−0.036 [−0.067, −0.008]	+0.018 [−0.013, +0.049]

Takeaway: Format SFT transfers output reliability to held-out graph families, but reasoning SFT has mixed numerical-evidence effects. Because both phases use gold graph targets, content imitation remains possible. Together with the newly generated larger graphs in W3, this addresses training-family overlap and direct topology familiarity without claiming that pretraining familiarity is eliminated. We will report all fold-seed runs and present post-training as a diagnostic case study, not a new training contribution.

W2 — Statistical power

We expanded the primary Sachs comparison from five to 30 paired realizations—independent data draws shared by the data and no-data conditions. Here, **data effect** means the data-use gain defined above. We report paired 95% bootstrap CIs and assign F1=0 to invalid outputs.

Model	Real names: data effect [95% CI]	Anonymous: data effect [95% CI]
Base (Qwen3-4B-Thinking-2507)	−0.183 [−0.234, −0.130]	+0.230 [+0.207, +0.252]
GPT-5-mini	−0.087 [−0.104, −0.069]	+0.039 [+0.017, +0.062]

Model	Real names: data effect [95% CI]	Anonymous: data effect [95% CI]
After format SFT	−0.091 [−0.143, −0.040]	+0.061 [+0.023, +0.100]
After format SFT + reasoning SFT	−0.089 [−0.168, −0.010]	+0.061 [+0.025, +0.096]

Takeaway: All eight paired sign-flip tests have $p \leq 0.0403$: adding data lowers F1 with real names but improves it anonymously in this Sachs slice. We treat the remaining five-realization cells as descriptive; this replication supports the prespecified contrast, not every benchmark cell.

W3 / Q2 / Q3 — Graph familiarity and scale

Familiarity. Chain-25 is a 25-node directed chain; Jungle-25 is a denser 25-node hierarchical graph. Both were generated for this evaluation. We also created five variants of the 20-node Child directed acyclic graph (DAG) that retain its variable names and each node's in- and out-degree while changing 12–16 of its 25 edges. On **disputed edges**, where the canonical and rewired Child graphs differ, the model after format SFT + reasoning SFT followed the rewired graph on 45.0% and the canonical graph on 43.6% with real names; anonymously, the rates were 41.4% and 43.6%. Thus, the supplied data did not create a consistent preference for the rewired structure. We use **name-mediated support** to mean performance enabled by variable names, whether it comes from exact benchmark recall or broader knowledge; these tests cannot distinguish those sources. These controls test unfamiliar topologies, but they do not provide a newly constructed named scientific domain or rule out broader pretraining exposure.

Scale. We evaluated the same real-name/anonymous and data/no-data comparisons on four larger graphs with 20 paired realizations per contrast. Exact repeated copies of a directed edge are collapsed before scoring because edge lists represent sets; other contract failures remain invalid. All LLM values below use this scoring rule and aggregate 20 runs per cell. ENCO uses the same 20 datasets and numerical budget. The table reports data-present LLM results after format SFT + reasoning SFT:

Graph	ENCO F1	LLM original: validity / F1	LLM anonymous: validity / F1
Child (20)	0.652	100% / 0.070	100% / 0.104
Chain-25	0.609	90% / 0.049	100% / 0.069
Jungle-25	0.703	95% / 0.044	100% / 0.095
Alarm (37)	0.677	60% / 0.022	90% / 0.031

We also report the data-use gain (F1 with data minus F1 without data) at these graphs, since that is the estimand our central claim rests on and it was missing from the table above:

Graph	Original-label gain	Anonymous-label gain
Child (20)	0.000	+0.005
Chain-25	0.000	0.000
Jungle-25	+0.003	0.000
Alarm (37)	−0.133	0.000

Takeaway: ENCO reaches F1=0.609–0.703 under the matched data budget, whereas LLM F1 remains 0.022–0.104 and its data-use gain is nearly zero except for a negative original-label Alarm effect. The latter partly reflects Alarm's 60% data-present validity because invalid outputs receive zero. High parse validity therefore does not imply successful recovery; these tests do not establish scalable LLM causal discovery.

Thank you again for the constructive suggestions. The requested experiments strengthen CausalMix's evaluation under held-out, repeated, and larger-graph settings. We will incorporate them and clarify the scope of our conclusions in the revision.

review

Official Review by Reviewer zDep 📅 24 Jun 2026, 05:30 (modified: 23 Jul 2026, 11:25)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer zDep

📄 Revisions (/revisions?id=KSaiIDm8CF)

Summary:

The paper primarily investigates whether causal graph recovery in LLM-based causal discovery relies on semantic priors or on the underlying data itself. To this end, the authors propose CAUSALMIX, a benchmark framework that systematically evaluates and analyzes model behavior by independently controlling variable semantics, data availability, and data representations. Experimentally, the paper compares several LLM families, including GPT, Qwen, and Llama, as well as classical causal discovery methods. The paper maintains a clear and focused research objective.

Contribution Type Check: Yes

Strengths:

1. Focused and well-motivated contribution: The paper concentrates on a fundamental question in causal graph recovery: what is the primary source of performance gains, semantic priors or numerical data? This issue extends beyond causal discovery and is broadly relevant to many LLM-based reasoning tasks. Therefore, the core contribution is meaningful and sufficiently important.
2. Well-designed benchmark framework: To rigorously evaluate the contribution of each information source, CAUSALMIX introduces four controlled experimental settings. The benchmark systematically isolates different factors and analyzes their effects through detailed F1-based comparisons. Overall, the evaluation protocol is carefully designed and logically structured.
3. Meaningful empirical findings: The experimental results suggest that many existing LLM-based causal discovery (LLM-CD) methods may rely more heavily on semantic priors than on genuine utilization of numerical evidence for causal graph recovery. This finding provides valuable insights into the limitations of current approaches and may inspire new directions for future research.

Weaknesses:

1. Limited persuasiveness of the graph structures used in the evaluation: The main experiments are conducted on graphs containing only 5–11 nodes. Such graphs are relatively small, with limited structural complexity and a constrained search space, yet the paper draws its primary conclusions from these settings. Although the authors release larger graphs as part of the benchmark, they do not perform sufficiently extensive evaluations on them. As a result, the propose of the paper is that current LLM-based causal discovery methods rely primarily on semantic information rather than numerical evidence remains insufficiently supported.
2. The verifier-based post-training approach is not fully convincing: As acknowledged by the authors, the training graphs and evaluation graphs are not completely disjoint. Furthermore, the paper lacks sufficiently strong ablation studies, adversarial evaluations, or controlled experiments to isolate the source of the observed improvements. Therefore, it is difficult to determine whether the reported gains truly stem from the proposed training strategy or from other confounding factors.
3. The experimental evidence is not sufficiently strong: Many of the reported performance differences are relatively small (e.g., F1 improvements of 0.05–0.10). It is unclear whether these differences reflect genuine performance gains or simply model variance and randomness. The paper lacks sufficiently thorough statistical analysis, additional experiments, or qualitative case studies to demonstrate that the observed improvements are meaningful and robust.
4. Insufficient diversity of evaluated models: As a benchmark paper, an important objective is to demonstrate the generality of its core findings across a broad range of LLMs. Therefore, evaluations should ideally cover both open-source and closed-source models with diverse architectures and capabilities. However, the current study only includes GPT, Qwen, and Llama families, which do not adequately represent either the classical models that have shaped the field or several state-of-the-art frontier models. As a result, it remains unclear whether the observed phenomenon is a general characteristic of modern LLMs or merely specific to the selected model families. Expanding the evaluation to include additional models would substantially strengthen the benchmark's conclusions and external validity.

Ethical Concerns: No or very minor ethics concerns only

Reproducibility: Yes

Reproducibility Comments:

The paper provides a relatively detailed description of the benchmark construction process, evaluation protocol, experimental settings, and metrics, which would allow readers to reproduce most of the methodology at a conceptual level.

Dataset Assessment: Partly

Dataset Comments:

Unable to access the URL

Limitations:

yes

Questions:

1. Evaluation on larger graphs: The main conclusions are derived from relatively small graphs. Could the authors provide additional experiments on larger graphs to further validate their findings? Such evaluations would help assess whether the observed patterns generalize to more complex causal structures and would strengthen the evidence supporting the paper's claims.
2. Indirectly ensuring train-test independence in post-training: The verifier-based post-training results may be affected by the overlap between training and evaluation graphs. Could the authors explore mechanisms to better ensure train-test independence, for example by controlling specific nodes, replacing semantically irrelevant variables, or modifying graph components while preserving overall structure? Such experiments would provide stronger evidence that the observed gains arise from the proposed training strategy rather than memorization.
3. Statistical significance analysis: Many reported improvements are relatively small in magnitude. Could the authors provide statistical significance tests, confidence intervals, or additional repeated runs to demonstrate that these improvements are attributable to the proposed method rather than random variation in model behavior?
4. Evidence supporting the claim that LLMs struggle to utilize numerical evidence: The propose of the paper is that current LLMs have difficulty leveraging numerical evidence for causal discovery. However, the paper provides limited direct evidence supporting this conclusion. Could the authors provide additional experiments, analyses, or references from prior literature to substantiate this claim? Otherwise, the motivation and interpretation of several findings may be less convincing.
5. Broader coverage of LLMs: The benchmark evaluation focuses primarily on GPT, Qwen, and Llama families. However, several influential frontier models, such as Gemini, Claude, and DeepSeek, are not included. As a benchmark paper, a broader model coverage would be important for demonstrating that the identified phenomenon is prevalent across the current LLM landscape. Without such evidence, it remains unclear whether the observed behavior is a general limitation of modern LLMs or merely a characteristic of the selected model families.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Paper Formatting Concerns:

One of the URLs provided in the paper could not be accessed during review, which made it difficult to verify the corresponding resource.

LLM Policy: Yes

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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Rebuttal by Authors

Edit ▾

Rebuttal by Authors 27 Jul 2026, 19:55 Program Chairs, Authors

Rebuttal:

Thank you for recognizing the importance of separating variable-name and numerical evidence. We added larger/newly generated graphs, held-out post-training ablations, a 30-run replication, and edge-level analyses; they show transferable format reliability but weak and unstable numerical recovery.

W1 / Q1 — Larger and more complex graphs

We ran the original/anonymous-label × data/no-data design on four 20–37-node graphs over 20 matched runs (4 column orders × 5 independent draws). The table reports data-present Qwen3-4B results after format and reasoning SFT. **Validity** means contract-compliant output at the expected graph size; acyclicity is separate, and invalid outputs receive F1=0. Exact duplicate edges are collapsed, while self-loops, unknown endpoints, malformed edges, and truncations remain invalid. All values below use this scoring rule and aggregate 20 runs per cell. ENCO uses the same 20 datasets and numerical budget.

Graph	ENCO F1	LLM original: validity / F1	LLM anonymous: validity / F1
Child (20)	0.652	100% / 0.070	100% / 0.104
Chain-25	0.609	90% / 0.049	100% / 0.069
Jungle-25	0.703	95% / 0.044	100% / 0.095
Alarm (37)	0.677	60% / 0.022	90% / 0.031

This does not contrast data-present against no-data at these graphs, which is the estimand our central claim rests on. That contrast (data-use gain, invalid=0) is:

Graph	Original-label gain	Anonymous-label gain
Child (20)	0.000	+0.005
Chain-25	0.000	0.000
Jungle-25	+0.003	0.000
Alarm (37)	−0.133	0.000

Takeaway: ENCO reaches F1=0.609–0.703 under the matched budget, confirming recoverable numerical signal. Even at 90–100% LLM validity on most cells, F1 remains 0.022–0.104 and data-use gains are approximately zero except for a negative original-label Alarm effect; that effect partly reflects Alarm's 60% data-present validity because invalid outputs receive zero. We do not claim successful large-graph LLM recovery.

W2 / Q2 — Post-training ablations and train–test independence

We agree that identifying the source of post-training gains requires explicit ablations. We therefore crossed three training stages—**Base**, format SFT, and format SFT + reasoning SFT—with a 2×2 information-source ablation: original versus anonymous labels and data versus no data. We evaluated this factorial design under leave-one-graph-out (LOGO) training, so every evaluated graph was excluded from both SFT phases. **Base** denotes Qwen3-4B-Thinking-2507 before our fine-tuning. Each of Cancer, Earthquake, Asia, and Sachs was held out in turn, and every fold was trained with seeds 42, 314, and 2718. Aggregate validity averages the four information-source conditions:

Held-out graph	Base	After format SFT	After format SFT + reasoning SFT
Asia	71.3%	100%	100%
Cancer	86.3%	100%	100%
Earthquake	76.3%	100%	100%
Sachs	73.8%	100%	100%

We define **data-use gain** as $F1(\text{with data}) - F1(\text{without data})$. The next table reports how reasoning SFT changed this gain relative to format SFT, with 95% bootstrap CIs over 60 paired predictions per graph and naming condition: the 20 data pairs above $\times 3$ training seeds.

Held-out graph	Real names: reasoning SFT effect [95% CI]	Anonymous: reasoning SFT effect [95% CI]
Asia	-0.071 [-0.116, -0.028]	+0.028 [+0.003, +0.054]
Cancer	+0.003 [-0.056, +0.061]	+0.004 [-0.056, +0.069]
Earthquake	-0.025 [-0.074, +0.027]	+0.037 [+0.012, +0.064]
Sachs	-0.036 [-0.067, -0.008]	+0.018 [-0.013, +0.049]

Candidate explanation	Ablation result	Conclusion
Output-contract learning	LOGO validity increases from 71.3%–86.3% for Base to 100% after format SFT.	Strongly supported.
Training-graph memorization	The validity gain persists when the evaluated graph is excluded from both SFT phases.	Not necessary for the validity gain.
Semantic dependence	LOGO names-only F1 is 0.498 versus 0.157 after format SFT and 0.489 versus 0.134 after reasoning SFT, original versus anonymous labels.	Strong name dependence; its pretraining source remains unresolved.
Numerical integration	Reasoning-SFT effects on data-use gain vary in sign across graphs and naming conditions.	Small and inconsistent.

Takeaway: These ablations identify transferable output reliability as the most robust gain and rule out evaluated-family overlap as necessary for it, but do not show consistent numerical-integration improvement. The rewired-Child ablation in Q4 adds an adversarial name–data conflict. Because both SFT phases contain gold graph targets, format learning, content imitation, and pretraining familiarity are not fully separated; we do not attribute these supervised-checkpoint gains to verifier optimization or claim complete mechanistic identification.

W3 / Q3 — Statistical and qualitative evidence

We expanded the primary Sachs comparison to 30 paired realizations—*independent data draws shared by the data and no-data conditions*. Here, **data effect** means the data-use gain defined above. We report paired 95% bootstrap CIs and assign $F1=0$ to invalid outputs.

Model	Real names: data effect [95% CI]	Anonymous: data effect [95% CI]
Base (Qwen3-4B-Thinking-2507)	-0.183 [-0.234, -0.130]	+0.230 [+0.207, +0.252]
GPT-5-mini	-0.087 [-0.104, -0.069]	+0.039 [+0.017, +0.062]
After format SFT	-0.091 [-0.143, -0.040]	+0.061 [+0.023, +0.100]
After format SFT + reasoning SFT	-0.089 [-0.168, -0.010]	+0.061 [+0.025, +0.096]

Takeaway: All eight paired sign-flip tests have $p \leq 0.0403$: adding data lowers F1 with real names but improves it anonymously in this Sachs slice. We treat the remaining five-realization cells as descriptive.

At the edge level, a **correction** changes an incorrect prediction to the true edge state, while a **regression** changes a correct prediction to an incorrect state. Across 60 valid real-name Sachs LOGO pairs, data corrected Jnk–P38 and Erk–P38 36 times each, but regressed PIP2–PKA and Mek–Plcg 31 times each. These cases show how local corrections and regressions can offset each other in graph-level F1.

Q4 — Direct numerical-evidence response

After format SFT + reasoning SFT, we compared predictions with and without data on every variable pair among paired-valid outputs. A **beneficial** change corrects an edge state, a **harmful** change breaks one, and **net benefit** is their rate difference. “Scale suite” denotes Child, Chain-25, Jungle-25, and Alarm; LOGO is defined in W2/Q2.

Evaluation	Paired-valid / total	Edges changed	Net benefit		
			Beneficial	Harmful	(percentage points)
Scale suite, original labels	69 / 80	18.7%	5.7%	11.4%	−5.7
Scale suite, anonymous	78 / 80	34.0%	5.0%	26.7%	−21.7
Strict leave-one-graph-out, real names	240 / 240	28.2%	11.4%	15.0%	−3.6
Strict leave-one-graph-out, anonymous	240 / 240	25.5%	9.8%	12.3%	−2.5
Sachs, 30 realizations, anonymous	30 / 30	42.2%	10.8%	25.5%	−14.7

Takeaway: As an adversarial ablation, five Child variants preserve names and node degrees while changing 12–16 of 25 edges. On disputed pairs, adding data reduced following of the rewired graph by 1.0 percentage point with real names and 7.9 anonymously, while increasing unresolved choices by 5.4 and 14.5 points. Across the broader paired-valid analyses, data change many local decisions but harmful changes often offset or exceed beneficial ones. We therefore replace “models ignore data” with **partial local responsiveness with unstable numerical-evidence integration**. We do not interpret a shuffled-data placebo whose prompts exceeded the context limit.

W4 / Q5 — Model coverage

The submission evaluates ten GPT-5/Llama/Qwen models, and the rebuttal adds a 30-run GPT-5-mini replication. During the rebuttal period, we also completed interim 10-run evaluations of Granite-3.2-8B and DeepSeek-V2-Lite-Chat (an actual DeepSeek-family checkpoint) on Child, Sachs, and larger graphs; 20-run replications are ongoing. We report these interim results only descriptively and do not use them for confidence intervals or small-effect claims. We will include the completed cross-family coverage, uncertainty estimates, and failure-mode analysis in the revised manuscript. Separately, three Ministral reasoning checkpoints produced 0% strict validity on the completed Child/Sachs/Ministral-Small-3.1-24B failed to load, and preliminary DeepSeek-R1-Distill-Llama-8B outputs were invalid. We treat these as protocol-compatibility failures, not causal-capability evidence; main inferential conclusions remain scoped to GPT-5/Llama.

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Formatting concerns (/open-data)

Dataset access

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On July 26, 2026, we verified through unauthenticated access that the dataset repository (<https://huggingface.co/datasets/causalbench/anonymous-data>) is public and un gated. We will make this access information explicit in the revised artifact documentation.

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